

Trainings ▾

General Information

Cummins Inc. recommends the use of fully-formulated antifreeze or coolant containing a precharge of supplemental coolant additive (SCA). The antifreeze or coolant **must** meet the specifications outlined in the Technology and Maintenance Council (TMC) Recommended Practice (RP) 329 (ethylene glycol) or Recommended Practice (RP) 330 (propylene glycol). The use of fully-formulated antifreeze or coolant significantly simplifies cooling system maintenance.

Copies of Technology and Maintenance Council (TMC) specifications can be obtained through Cummins Inc., or by contacting:

Technology and Maintenance Council

American Trucking Association

2200 Mill Road

Alexandria, VA 33314-5388

Phone: (703) 838-1763

Fax (703) 836-6070

Fully-formulated antifreeze contains balanced amounts of antifreeze, SCA, and buffering compounds, but does **not** contain 50 percent water. Fully-formulated coolant contains balanced amounts of antifreeze, SCA, and buffering compounds already premixed 50/50 with deionized water.

Alternative maintenance practices for cooling systems can be found in Cummins® Coolant Requirements and Maintenance, Bulletin 3666132.

Specifications

Automotive Applications

Coolant Capacity (engine only)

With EGR	33.1 liters [35 qt]
Without EGR	24 liters [25 qt]

Standard Modulating Thermostat

Temperature Range	82 to 93°C [180 to 200°F]
Maximum Coolant Pressure (exclusive of pressure cap - closed thermostat at the maximum no-load governed speed)	
ISX CM871 (with EGR)	434 kPa [63psi]
ISX CM870 (with EGR)	400 kPa [38 psi]
ISX Without EGR (without EGR)	227 kPa [33 psi]
Coolant Alarm Activation Temperature	
With EGR - Ratings Below 565 Horsepower	107°C [225°F]
With EGR - Ratings of 565/600 Horsepower	110°C [230°F]
Without EGR	110°C [230°F]
Maximum Allowable Top Tank Temperature	
Ratings Below 565 Horsepower with EGR and without EGR Engines	107°C [225°F]
Ratings of 565/600 Horsepower with EGR only	110°C [230°F]
Minimum Recommended Top Tank Temperature	
Minimum Temperature	70°C [160°F]
Minimum Allowable Draw Down	
With EGR	11 Percent
Without EGR	2.4 liters [2.5 qt] or 10 Percent of System Capacity (whichever is greater)
Minimum Recommended Pressure Cap	
With EGR	103 kPa [15 psi]
Without EGR	50 kPa [7 psi]
Minimum Fill Rate	
Without Low-Level Alarm	19 liters/min [5 gpm]
Maximum Deaeration Time	
Maximum Time	25 minutes
Fan-on Coolant Temperature	
With EGR	99°C [210°F]

Without EGR	99°C [210°F]
Fan-on Intake Air Temperature	
With EGR	93°C [200°F]
Without EGR	88°C [190°F]
Shutter Opening Temperature	
Coolant - with EGR	96°C [205°F]
Coolant - without EGR	85°C [185°F]
Intake Air - with EGR	104°C [220°F]
Intake Air - without EGR	66°C [150°F]
Winterfronts	
Air passage area	774 cm ² [120 in ²]
Industrial and Generator Drive	
Coolant Capacity	
Engine Only	24 liters [25 qt]
Standard Modulating Thermostat	
Temperature Range	82 to 93°C [180 to 200°F]
Maximum Coolant Pressure (Exclusive of pressure cap - closed thermostat at the maximum no-load governed speed)	
All ratings	227 kPa [33 psi]
Coolant Alarm Activation Temperature	
Ratings Below 565 Horsepower	107°C [225°F]
Ratings of 565/600 Horsepower	110°C [230°F]
Maximum Allowable Top Tank Temperature	
Ratings Below 565 Horsepower	107°C [225°F]
Ratings of 565/600 Horsepower	110°C [230°F]
Power Generation, Standby	110°C [230°F]
Power Generation, Prime	104°C [220°F]
Minimum Recommended Top Tank Temperature	

Minimum Temperature	70°C [160°F]
Minimum Allowable Draw Down	
Minimum Allowable Draw Down	2.4 liters [2.5 qt] or 10 Percent of System Capacity (whichever is greater)
Minimum Recommended Pressure Cap	
Industrial	103 kPa [15 psi]
Power Generation	69 kPa [10 psi]
Minimum Fill Rate	
Without low-level alarm	19 liters/min [5 gpm]
Maximum Deaeration	
Time	25 minutes
Fan-on Coolant Temperature	
Temperature	95°C [203°F]
Fan-on Intake Air Temperature - Industrial Only	
Temperature	88°C [190°F]
Shutter Opening Temperature - Industrial Only	
Coolant	85°C [185°F]
Intake	60°C [140°F]